

KAGOSHIMA, Japan

WMO: 478270

Lat: 31.5550N

Lon: 130.5481E

Elev: 32

StdP: 100.95

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.6	2.7	-6.7	2.1	6.8	-5.3	2.4	6.5	9.2	9.2	8.5	9.1	3.5	340	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	33.6	26.0	32.7	25.8	31.9	25.6	27.2	31.3	26.7	30.7	26.3	30.2	4.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.6	29.3	25.6	21.0	29.1	25.2	20.4	28.9	86.1	31.4	84.0	30.9	82.2	30.2	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	7.7	6.8	-0.7	35.2	1.3	0.8	-1.6	35.8	-2.3	36.2	-3.1	36.7	-4.0	37.3		
			-2.5	27.8	1.1	0.6	-3.3	28.2	-4.0	28.6	-4.6	29.0	-5.4	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.1	8.7	10.1	12.9	17.2	21.3	24.3	28.5	29.2	26.6	22.0	16.3	11.0
	DBStd	7.51	2.93	3.34	3.04	2.71	1.85	2.15	1.74	1.40	2.14	2.72	3.32	3.29
	HDD10.0	132	60	36	8	0	0	0	0	0	0	0	1	27
	HDD18.3	1062	298	231	170	52	2	0	0	0	0	4	78	228
	CDD10.0	3441	21	38	98	216	351	428	575	597	499	371	189	58
	CDD18.3	1329	0	0	1	18	94	178	317	338	249	117	17	1
	CDH23.3	12253	0	0	1	31	316	1092	3666	4162	2393	567	24	0
	CDH26.7	4299	0	0	0	1	23	217	1452	1800	734	71	1	0
Wind	WSAvg	3.3	3.4	3.5	3.6	3.4	3.1	3.0	3.2	3.3	3.3	3.3	3.2	3.4
Precipitation	PrecAvg	2297	83	105	160	217	236	463	312	218	221	106	98	82
	PrecMax	4025	201	250	351	453	559	858	1055	630	532	361	216	245
	PrecMin	1523	9	31	33	41	55	197	17	23	29	11	26	8
	PrecStd	455	44	55	74	104	100	182	179	130	125	82	51	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.8	20.8	22.5	25.8	28.5	31.9	34.5	34.9	33.1	29.9	25.6	21.1
		MCWB	15.3	17.1	16.4	18.7	19.7	25.6	26.1	26.3	25.2	23.4	20.2	16.5
	2%	DB	16.7	18.9	20.7	24.1	27.2	30.2	33.5	33.8	31.9	28.5	23.9	19.2
		MCWB	13.2	15.5	15.4	17.7	19.7	24.9	25.9	26.2	24.9	22.3	18.8	15.5
	5%	DB	15.2	17.2	19.4	22.9	26.1	28.8	32.6	33.0	31.0	27.3	22.7	17.9
		MCWB	11.6	13.7	14.7	17.0	19.5	24.0	25.8	25.9	24.5	21.5	18.2	13.9
	10%	DB	13.9	15.7	18.1	21.8	25.1	27.8	31.8	32.1	30.1	26.2	21.6	16.5
		MCWB	10.4	12.1	13.9	16.6	19.1	23.3	25.7	25.7	24.1	20.8	17.3	12.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.5	18.1	18.7	21.6	23.7	26.3	27.7	28.1	26.9	25.2	22.1	18.4
		MCDB	18.1	19.7	20.5	23.8	26.2	29.5	31.7	32.3	30.5	28.0	24.4	19.5
	2%	WB	14.4	16.3	17.2	20.1	22.6	25.7	27.1	27.3	26.2	24.1	20.5	16.4
		MCDB	16.0	18.3	19.1	22.1	24.7	28.8	31.2	31.5	29.7	27.1	22.6	18.4
	5%	WB	12.6	14.6	16.0	18.9	21.7	25.2	26.6	26.9	25.7	23.2	19.3	14.9
		MCDB	14.5	16.7	18.4	21.2	23.8	28.1	30.6	31.0	29.2	26.3	21.6	17.3
	10%	WB	10.7	12.8	14.8	17.9	20.7	24.6	26.2	26.5	25.3	21.9	18.1	13.4
		MCDB	13.4	15.1	17.5	20.4	23.4	27.2	30.1	30.5	28.7	25.0	20.9	16.1
Mean Daily Temperature Range	5% DB	MDBR	7.4	7.8	8.1	8.0	7.4	5.5	5.8	5.9	6.2	7.0	7.4	7.6
		MCDBR	9.0	8.6	9.3	9.3	8.9	6.6	7.0	6.9	6.8	7.3	7.5	8.4
		MCWBR	6.4	6.3	5.9	5.0	4.3	2.8	2.4	2.4	2.8	3.6	4.4	5.7
	5% WB	MCDBR	7.6	7.4	7.5	6.7	5.8	5.4	6.2	6.1	5.7	5.6	5.5	7.0
		MCWBR	6.9	6.5	6.1	4.9	4.2	2.7	2.5	2.4	2.5	3.4	4.4	5.9
Clear-Sky Solar Irradiance	taub	0.428	0.465	0.521	0.539	0.546	0.546	0.527	0.517	0.502	0.469	0.431	0.419	
	taud	2.102	2.031	1.892	1.902	1.951	2.040	2.127	2.155	2.150	2.159	2.188	2.135	
	Ebn at Noon	768	779	764	770	767	764	776	779	774	768	762	753	
	Edh at Noon	131	155	190	196	188	171	156	150	146	135	120	121	
All-Sky Solar Radiation	RadAvg	2.39	3.04	4.07	4.78	4.96	4.05	5.35	5.40	4.50	3.75	2.78	2.26	
	RadStd	0.22	0.27	0.28	0.48	0.56	0.64	0.60	0.38	0.47	0.42	0.26	0.22	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47)	Regional (20 neighbors)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page